Routine motor inspection

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Before work, identify the motor, driven load, operating duty, and recorded baseline. Inspect the frame for impact, corrosion, loose guards, oil or water ingress, and blocked ventilation. Record ambient temperature, load, sound, vibration, and bearing housing temperature at repeatable points. Compare readings with the equipment record and site limits; this guide does not define universal acceptance limits.

Preventive maintenance intervals must follow the site schedule and motor supplier requirements. Keep a dated log of observations, measurements, corrective work, and parts fitted. A change from the established baseline warrants investigation even when the motor still runs.

Cooling and overheating

Overheating can result from blocked cooling passages, excessive ambient temperature, overload, poor ventilation, unbalanced supply, or bearing friction. First review load and operating history and compare temperature readings with the established baseline and site limits. With the equipment safely isolated, inspect external cooling fins, fan cover, and air passages for dust or debris. Clean only by an approved method that will not drive contamination into the motor. Confirm the fan and shroud are intact and unobstructed.

Do not touch a hot housing to estimate temperature. Use a suitable calibrated instrument from a safe position. If temperature continues to rise, trips recur, or the cause is uncertain, stop and refer to a qualified electrical or mechanical technician.

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Lubrication and bearings

Bearing noise, rising housing temperature, or vibration may indicate lubrication problems, contamination, misalignment, or wear. Record the bearing location and operating condition. Check the maintenance record for lubricant type, quantity, and service date. Use only the specified lubricant and approved service method; incompatible greases can damage bearings. Do not add lubricant solely because a bearing is noisy.

Inspect for leakage, damaged seals, and contamination after isolation. A rough bearing, persistent abnormal noise, or repeat temperature rise requires qualified assessment. Bearing replacement and internal motor work are outside this guide.

Belt inspection and alignment

A belt drive can produce vibration, heat, and premature wear when tension or alignment is incorrect. After safe shutdown and isolation, inspect belt surfaces for glazing, cracking, fraying, contamination, and uneven wear. Check pulley grooves and mounting for visible damage. Use an appropriate alignment tool and the equipment specification to assess pulley alignment. Use the drive maker's specified tension method; do not estimate tension by hand.

Replace matched belt sets together where the drive design requires it. Refit guards before operation. If alignment or tension cannot be verified to specification, refer the work to qualified maintenance staff.

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Electrical connections

Discoloration, odor, repeated overload trips, or an unexpectedly hot terminal may indicate an electrical fault. Visual inspection alone cannot confirm a safe or sound connection. Only qualified personnel using approved isolation, verification, and torque procedures may inspect or service electrical terminals. Compare supply conditions and recorded motor current with the equipment nameplate and site limits. Do not open an energized enclosure. Escalate recurring trips, visible arcing, damaged insulation, or uncertain measurements.

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